K.A.R.M.A · PRESENTATION

KARMA Presentation Speaker Guide

How to present each of the 25 slides in `presentation/index.html`: what is on screen, what to say and what to click.

Team K.A.R.M.A · The Oxford Evening Polytechnic · Semester VI Diploma Major Project

Before you start

The deck has 25 slides and runs 18–22 minutes; aim for 20, then questions. Open `presentation\index.html` in Chrome or Edge and press **F** for full screen. →, **Space** or a clicker moves forward; ← goes back.

- **Build slides:** slides 3, 4, 8, 9, 14, 15, 16, 18 and 19 reveal step by step. Keep pressing → until the slide is complete before moving on.
- **Live slide 13:** click the command line to type. Press **Esc** before pressing Space, or Space types a space instead of moving on.
- **KARMA running:** start the server and press **Reset System** before you begin, so the live demo starts clean.
- **Split between speakers (suggested):** Part 1 one speaker, Part 2 the second, Part 3 the lead with the live demo, Part 4 the fourth. Hand over with one sentence that links the parts.

Part	Slides	Time	Goal
1. The problem	1–7	4 min	Make the examiner feel the risk
2. The idea	8–13	5 min	Show how a honeypot catches attackers
3. KARMA inside	14–19	6 min	The demo story and how it is built
4. Proof and future	20–25	5 min	Security, tests, limits, conclusion

Part 1 — The problem (slides 1–7, about 4 minutes)

Goal: by slide 7 the examiner should agree that walls are not enough and traps are needed.

#	Slide	On screen	What to say	Do
1	Cold open	A terminal types an SSH login, `whoami`, `cat /etc/passwd`; then "Someone just broke in."	Stay silent while it types. Then: "Someone just broke in. Not into our server — into our trap."	Wait for the typing to finish
2	Cover	"The attack returns." with the live map video; project name and college	"We are Team KARMA. KARMA is a honeypot platform: it lures attackers into fake servers, reads every move and locks them out."	Introduce the team in one line
3	Breached since 2024	Five company names light up one by one with real headlines: Salesforce, Coupang, Co-op, JLR, Canvas	One line per press: "1 billion records claimed from Salesforce customers..." End: "These companies had firewalls."	Press → five times
4	And here in India	Star Health, Tata Technologies, CERT-In 29.44 lakh incidents	"This is not somewhere else. 31 million Star Health policyholders' records were posted online. CERT-In handled 29.44 lakh incidents in 2025."	Press → three times
5	241 days	A grid fills with 241 red squares	"On average an intruder is inside for 241 days before anyone notices. Each square is a day they walk around unseen. Average cost: $4.44 million (IBM 2025)."	Let the counter reach 241
6	Attackers bring keys	22% stolen credentials, 44% ransomware, 30% third parties	"A firewall checks the door. But 22% of breaches start with a valid stolen password — the firewall waves it through."	Point at 22%
7	Stop building walls. Lay traps.	Firewall vs honeypot comparison	"A honeypot is a decoy that looks more valuable than the real server. Nobody has a reason to touch it, so anyone who does is an intruder — almost no false positives."	Hand over to Part 2

Handover line: "So how does a trap actually catch someone? ..."

Part 2 — The idea (slides 8–13, about 5 minutes)

Goal: explain the honeypot loop once, clearly, then let the examiner try it on slide 13.

#	Slide	On screen	What to say	Do
8	Lure → Trap → Profile → Contain	A diagram: the attacker aims at the real server, the path bends into the KARMA decoy, a tag shows T1059 risk 90, then BLOCKED	Lure: "KARMA runs fake services that look real." Trap: "The attacker reaches the decoy first; the real server is never touched." Profile: "Every command is logged, mapped to MITRE ATT&CK and scored 0–100." Contain: "Cross 75 and KARMA quarantines them."	Press → three times (or drag the slider)
9	KARMA	The word KARMA opens into its letters	"In security, karma is an attack that comes back to its sender. KARMA turns your own network into the trap."	Press → once to open the letters

#	Slide	On screen	What to say	Do
10	Objectives	Five numbered goals: Lure, Record, Classify, Contain, Explain; out-of-scope line	Read the five bold words, one sentence each. Mention scope: "We did not aim to guard a real production network or use machine learning."	—
11	Others do one part	Table: firewall, Cowrie, T-Pot, Wazuh vs KARMA	"Cowrie only logs. T-Pot needs a Linux server running Docker. Wazuh has no decoys. KARMA does the whole loop on one laptop."	Point at the amber KARMA row
12	Live attack map	Real recording of the visualizer	"This is a real recording. Every attack starts on course for production; the honeypot pulls it in."	Let the video play 5–10 seconds
13	Type what an attacker types	A command line, what the attacker sees, and KARMA's verdict	"Try it." Type or click `whoami` (T1059, risk 90, quarantines), then `ls -la` (logged only), then `' OR 1=1 --` (T1190 SQL injection). "These are the same rules KARMA's engine uses."	Click the line to type; **Esc** before moving on

Handover line: "Now let's watch a full attack from start to finish."

Part 3 — KARMA inside (slides 14–19, about 6 minutes)

Goal: tell one attack story, show the real screens, then explain how one keystroke flows through the system. If you run the live demo, do it right after slide 14.

#	Slide	On screen	What to say	Do
14	Steal the keys. Trip the alarm.	Six stages: Recon, Web recon, Break-in, Loot, Use the key, Contained	One sentence per press, as shown under the cards. End with: "In Contain mode only high-risk acts block; Observe mode never blocks, so we can rehearse."	Press → five times; then switch to the live demo if planned
—	Live demo (optional, 3 min)	Attacker laptop + KARMA dashboard	Run the attack script from the Decoys and Attacks Guide. On the dashboard open ▶ **Replay** and 📄 **Report** for the attacker	Press **Reset System** before; return with the slide tab
15	One console for the whole hunt	Real screenshots: dashboard, MITRE heatmap, honeytoken vault	Dashboard: "Every probe appears the second it happens." MITRE: "Brute force becomes T1110, SQL injection T1190, a stolen key T1078." Honeytokens: "Fake keys planted to be stolen; using one sets off the alarm."	Press → twice
16	It watches, explains, and calls you	Decoy control, AI copilot, alerts	"Five decoys, nothing runs on the PC." "Ask the AI copilot 'who is 192.168.137.5?' — it works even offline." "One phone alert when an attack starts, no spam."	Press → twice

#	Slide	On screen	What to say	Do
17	And everything around it	Six cards	Pick three: phishing analyzer, decoy documents, the internet tunnel. "It also packs into a single .exe."	—
18	One keystroke, end to end	A packet moves through Attacker → Paramiko → Virtual shell → MITRE mapper → SQLite → FastAPI	Read the line under each step. Key point: "The virtual shell answers 'root' from a fake file system; the command never touches Windows."	Press → five times
19	Five layers, one Python process	Stacked plates: Interface, Server, Brain, Traps, Memory	One sentence per layer: Vanilla JS + Chart.js + Leaflet; FastAPI + WebSockets; MITRE mapper + threat score + AI; Paramiko and raw sockets; SQLite + CSV audit.	Press → four times

Handover line: "A honeypot invites attackers in, so the first question is: is KARMA itself safe?"

Part 4 — Proof and future (slides 20–25, about 5 minutes)

Goal: answer the examiner's doubts before they ask, then close strongly.

#	Slide	On screen	What to say	Do
20	Built to be attacked	Four cards: Nothing runs, Console stays private, Floods are capped, Secrets stay hidden	"The shell is simulated — no programs, pipes or host files. The dashboard answers only the KARMA PC and needs a 43-character token. 10 logins and 600 requests a minute per IP. Passwords are redacted from logs."	—
21	Fourteen tests. All passed.	TC-01 to TC-14, each PASS	"These are the 14 system test cases from our report, run on the live platform." Explain two: TC-07 (auto-quarantine, block and unblock) and TC-13 (dashboard refused from outside). "Automated checks also run on Windows and Linux for every change."	Point at TC-07 and TC-13
22	KARMA, counted	5 decoys, 6 MITRE techniques, 14 test cases, 0 commands run on the host	Read each number with its line. Stress the 0.	Let the counters finish
23	Where it goes next	Today's limits vs future scope	"Honest limits: one laptop, rule-based detection, approximate geolocation." Then two future items: "firewall integration and distributed sensors."	—
24	They came for our data. We got theirs.	Conclusion	"KARMA shows one ordinary laptop can lure, profile and contain an attacker in real time. Every attacker who touches the trap leaves their IP, tools, commands and intent behind."	Pause after the headline

#	Slide	On screen	What to say	Do
25	Thank you. Questions?	Team photos and roles	"Thank you. The honeypot is still running — ask us anything."	Leave this slide up during questions

During questions: use the KARMA Viva Q&A (KARMA_Viva_QA.pdf). If asked to show something, switch to the dashboard: ▶ **Replay** for an attacker's session, 📄 **Report** for the incident report.